Portfolio Sample · n8n Silent-Failure Diagnostic & Hardening Plan

> Client: an agency with an n8n lead workflow that "works, then silently stops" (fictional demo client). Ranked root-cause analysis (most likely first, each tied to their exact symptoms), a confirmation check + fix per cause, a hardening plan, and an "if you can only do 3 things today" list. Diagnose/fix-type delivery.
>
> Delivered through a production → independent QA → fix → re-QA pipeline. **Final QA verdict: PASS.**

Part 1 · Client Brief & Delivery Spec

Delivery Spec — n8n Silent Failure Diagnostic & Hardening

Goal

Provide a diagnostic framework and actionable hardening plan for a self-hosted n8n workflow that intermittently stops processing inbound leads without surfacing errors, requiring manual restart to resume operation.

Scope

In Scope	Out of Scope
Root cause analysis for the specific workflow: webhook → set fields → OpenAI → Gmail → Google Sheets	Infrastructure migration (e.g., moving off self-hosted VPS)
Diagnostic checks executable by a non-technical agency owner	Writing custom n8n nodes or code
Configuration changes within n8n and connected services	Debugging the website form that sends the webhook
Error workflow/alerting setup in n8n	Performance optimization beyond reliability
Retry, timeout, and credential handling configurations	General n8n platform education

Deliverables

D1. Ranked Root Cause Analysis

Ordered list of likely causes specific to: self-hosted n8n VPS, OpenAI node, Gmail node, Google Sheets node, ~30-40 executions/day, pattern of silent failure with manual restart temporarily resolving.

Each entry must include:

- **Why it fits their symptoms** (tied to: "stops silently," "no error in inbox," "last execution failed but can't tell why," "sometimes OpenAI node," "restart fixes for a few days")
- **Specific confirmation check** (n8n UI path, command, or external service check)
- **Fix** (configuration change, setting, or operational adjustment)

D2. Hardening Plan

Changes to prevent silent failures going forward, covering:

- Error workflow / alerting (default: basic error workflow triggering notification)
- Retries (node-level and workflow-level)
- Timeouts (execution and node-specific)
- Credential handling (refresh, expiry, rate limits)
- Queue/execution settings relevant to self-hosted VPS resource constraints

D3. Prioritized Action List

"If you can only do 3 things today" — the highest-impact interventions from D1+D2, ranked by likelihood of stopping the silent failure pattern with least effort.

Acceptance Criteria

ID	Criterion	Traces To Client Statement
AC1	Root causes ranked by likelihood, not presented as equally probable	"most likely first" in deliverable expected #1
AC2	Each root cause explicitly connects to their specific symptoms	"with *why* each fits their symptoms (not generic advice)"
AC3	Checks are n8n-UI-specific or service-specific, not generic "check logs"	"No generic 'check your logs' fluff"
AC4	Stack-specific throughout: n8n self-hosted, OpenAI node, Gmail, Sheets	"specific to their stack"
AC5	Hardening plan addresses silent failure specifically, not just general best practices	"not fail silently again"
AC6	"If you can only do 3 things today" is extractable as standalone prioritized list	Deliverable expected #4
AC7	Output in English	"English"
AC8	No reliance on client-provided error screenshots or additional data gathering	"They pasted no error text (say 'I'll send a screenshot if needed')" — spec must be actionable without this

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Edge Cases

Scenario	Handling
Multiple concurrent root causes	D1 ranking should help isolate; D2 hardening addresses systemic resilience regardless of specific cause

Scenario	Handling
VPS resource exhaustion masks as node failure	Include in D1; "sometimes it's the OpenAI node, sometimes it looks fine" suggests resource contention or OOM kill
OpenAI rate limits vs. OpenAI timeouts	Distinguish in D1: rate limit = HTTP 429 with headers; timeout = no response; both fit "stops silently" depending on n8n error handling config
Gmail OAuth refresh token expiry	Include in D1: fits "restart fixes it" if manual restart triggers re-auth, plus "no error in inbox" if error workflow not configured
Google Sheets API quota (100 requests/100 seconds/user)	Include in D1: at 30-40/day normally low, but burst + retries could trigger; fits intermittent pattern
Self-hosted n8n default: no error workflow configured	D2 must explicitly address; default n8n install does not enable error workflows (default)
Client cannot implement all hardening at once	D3 provides fallback; D2 should indicate which items are sequential dependencies vs. independent

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Reasonable Defaults (Client Silent)

Item	Default	Rationale
Error workflow notification destination	Email to workflow owner (default)	Client said "no error in my inbox" — implies email is their checked channel; n8n default is no notification
Retry count for failed nodes	2 retries with exponential backoff (default)	Industry standard; client specified no preference
Execution timeout ceiling	5 minutes (default)	OpenAI generation + Gmail send + Sheets append; generous enough for normal operation, tight enough to catch hangs
Error workflow existence	Must be explicitly created (default: none exists)	n8n default behavior; client symptom confirms this

Part 2 · Final Deliverable

n8n Silent Failure Diagnostic & Hardening Plan

D1. Ranked Root Cause Analysis

#1: No Error Workflow Configured (Systemic Blind Spot — Reveals Silence, Not Failure Itself)

Why it fits: n8n self-hosted defaults to *no* error workflow. When any node fails, the execution stops with no external notification. "No error in my inbox" confirms absence. "Last execution failed but can't tell why" matches executions showing "Error" in n8n UI but no proactive alert. "Restart fixes it" is manual recovery from unalerted failures. *Note: This explains why failures are invisible, not why they occur — it amplifies every other root cause into a "silent" failure.*

Confirmation check: n8n UI → Settings → Error Workflow. If dropdown shows "—" or no workflow selected, this is confirmed. Also: Executions (left sidebar) → filter by "Error" — if errors exist but you weren't notified, confirmed.

Fix: Create dedicated error workflow (see D2). Set as default in Settings → Error Workflow. Add email notification node to error workflow.

#2: OpenAI Node Failure — Timeout Hang vs. Rate Limit (429)

Why it fits: Two distinct OpenAI failures fit "sometimes it's the OpenAI node" with different signatures. Default n8n node timeout is 0 (unlimited), so OpenAI API can hang without response on overloaded models or degraded routes — execution stays "Running" indefinitely, blocking subsequent nodes. *Separately*, OpenAI rate limits (HTTP 429) return an explicit error that, without error workflow, also appears silent. Both explain manual restart as temporary fix; the distinction matters for confirmation and fix.

Symptom	Timeout Hang	Rate Limit (429)
Execution status	Stuck "Running"	"Error" (invisible without error workflow)
Client observation	"Sometimes it's the OpenAI node" then nothing	Intermittent failure, works after restart (backoff period elapsed)
Root cause	OpenAI unresponsive; n8n waits forever	Exceeded tokens-per-minute or requests-per-minute quota

Confirmation check — Timeout: Executions → identify stuck "Running" executions older than 10 minutes. Click execution → inspect if paused at OpenAI node. Settings → Execution timeout — if 0 or blank, confirmed.

Confirmation check — Rate limit: Executions → filter by "Error" → click failed execution → OpenAI node error details. Look for `429` status code or `RateLimitError` / `RequestsPerMinuteError`. Check OpenAI Dashboard → Usage → Rate limits for your tier.

Fix — Timeout: Settings → Execution timeout: **300 seconds** (5 min). OpenAI node options → Options → Timeout: **60 seconds**. Enable "Continue On Fail" with fallback branch for timeout handling.

Fix — Rate limit: OpenAI node → Retry: **2** with exponential backoff (1s, 2s, 4s). If 429 persists, upgrade OpenAI usage tier or implement custom delay node honoring `Retry-After` header from error response. Add error branch on OpenAI node to log 429s to your error workflow.

#3: Gmail OAuth Refresh Token Silent Expiry

Why it fits: Google OAuth refresh tokens expire after 6 months of inactivity or if consent revoked. n8n may not surface this as a clear error — execution fails at Gmail node with cryptic auth message. "Restart fixes it" if restart coincides with re-auth flow or token refresh edge case. "No error in my inbox" because error workflow absent.

Confirmation check: Credentials → Gmail → Test credential. If fails with "invalid_grant" or similar, confirmed. Google Cloud Console → APIs & Services → Credentials → OAuth 2.0 Client IDs → check last used date.

Fix: Re-authorize Gmail credential. In Google Cloud Console, set publishing status to "In production" (not Testing) to avoid 7-day test token expiry. Enable "Continue On Fail" on Gmail node with error branch logging to Sheets or separate notification.

#4: VPS OOM Kill or Resource Exhaustion

Why it fits: Self-hosted VPS with limited RAM. OpenAI responses + concurrent executions can spike memory. Linux OOM killer terminates n8n process without clean error logging. "Sometimes looks fine, sometimes OpenAI node" fits non-deterministic memory pressure. Manual restart restores service. Intermittent pattern matches load variation.

Confirmation check: SSH to VPS → `dmesg | grep -i "killed process"` or `journalctl -k | grep "Out of memory"`. Check `free -h` during active period. n8n Settings → Executions → check if "Running" executions spike before failures.

Fix: `docker run` or `docker-compose`: add `--memory="1.5g"` limit and swap. n8n Settings → Concurrency: limit to **2 executions** (default unlimited). Enable "Save Execution Progress" to reduce memory per execution.

#5: Google Sheets API Quota Burst + Retry Amplification

Why it fits: 100 requests/100 seconds/user quota. At 30-40/day normally safe, but burst scenarios (batch leads, retry storms) trigger 429 errors. Without error workflow, 429s silent. n8n default retry may amplify quota violation.

Confirmation check: Google Cloud Console → APIs & Services → Google Sheets API → Quotas → check "Read requests per 100 seconds" for 429 spikes. n8n Executions → filter by error → inspect Sheets node error details.

Fix: Google Sheets node → Options → Batch operations where possible. Reduce node-level retries to **1** for Sheets. Add 2-second delay node before Sheets if burst pattern detected.

D2. Hardening Plan

Error Workflow & Alerting (Critical: Addresses AC5 Silent Failure)

Element	Configuration
Error workflow creation	New workflow → trigger: "Error Trigger" → add "Send Email" node
Email recipient	Workflow owner address (your checked inbox)
Email content	`Execution ID: {{ $execution.id }}`, `Workflow: {{ $execution.workflowId }}`, `Error: {{ $json.error.message }}`, `Failed node: {{ $json.error.node.name }}`
Default assignment	Settings → Error Workflow → select created workflow

Operational note: Test by creating throwaway workflow with "Code" node executing `throw new Error("test")`.

Retry Configuration

Level	Setting	Rationale
Workflow-level	Settings → Error workflow (above)	Catches unhandled node failures
Node-level OpenAI	Retry: **2**, Wait between: **exponential** (1s, 2s, 4s), Max wait: **8s**	Handles transient OpenAI degradation and 429 recovery without hammering
Node-level Gmail	Retry: **1**, Wait: **2s**	OAuth transient recovery; avoids token lock
Node-level Google Sheets	Retry: **1**, Wait: **3s**	Quota recovery without amplification

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Timeout Configuration

Scope	Setting	Location
Global execution	**300 seconds** (5 min)	Settings → Execution timeout
OpenAI node	**60 seconds**	Node → Options → Timeout
Gmail node	**30 seconds**	Node → Options → Timeout
Google Sheets node	**30 seconds**	Node → Options → Timeout

Critical: Any timeout must pair with "Continue On Fail" + error branch to prevent silent drop.

Credential Handling

Service	Action	Frequency
Gmail	Verify "In production" status in Google Cloud Console	One-time
Gmail	Set calendar reminder to re-auth at 5 months	Every 5 months preventive
OpenAI	Verify API key has usage tier with rate limit headroom	Check monthly
All	Credentials → Test credential after any n8n version update	Per update

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Queue/Execution Settings (VPS Resource Constraints)

Setting	Value	Path
Concurrency	**2** max	Settings → Concurrency
Save execution progress	**Enabled**	Settings → Save execution progress
Save manual executions	**Disabled** (reduce storage)	Settings → Save manual executions
Prune executions	**30 days**	Settings → Prune old executions

Docker/Compose addition:

```
services:
  n8n:
    mem_limit: 1.5g
    memswap_limit: 2g
    restart: unless-stopped
```

Dependency Sequencing

Dependency	Explanation
Sequential: Error workflow must be created and set as default BEFORE timeout changes	Timeouts without alerting reintroduce silence — a hung execution killed by timeout produces an error that only surfaces if error workflow exists
Independent: Retry settings, credential refreshes, queue/concurrency limits	Can be applied in any order once error workflow is active

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D3. Prioritized Action List: "If You Can Only Do 3 Things Today"

Priority	Action	Time	Impact
1	**Create error workflow with email alert; set as default**	15 min	Eliminates silent failure permanently — every future error surfaces to your inbox
2	**Set execution timeout to 300s; OpenAI node timeout to 60s**	5 min	Stops hung executions blocking all processing; explains "restart fixes it"
3	**Test Gmail credential; re-authorize if any doubt; set to "In production"**	10 min	Fixes auth-related intermittent failure; token expiry is time-bomb at 6 months

Cumulative time: 30 minutes. **Result:** No future failure remains silent; hung executions self-terminate; OAuth expiry prevented.

Implement D3 immediately. Return to D1 diagnostic checks if failures persist after hardening. D2 remaining items can follow sequentially: error workflow is prerequisite for timeout changes; retry, credential, and queue settings are independent and can follow in any order.

Part 3 · Independent QA Report (final round)

Prior QA Issue Disposition

#	Issue	Status	Evidence
1	OpenAI rate limit (429) not distinguished from timeout	**RESOLVED**	D1 #2 now contains a comparison table (Timeout Hang vs. Rate Limit 429) with separate confirmation checks ("filter by 'Error' → look for `429` / `RateLimitError`" vs. stuck "Running") and separate fixes (429 fix: retries with backoff, `Retry-After` header, error branch)
2	D2 dependency sequencing under-developed	**RESOLVED**	New "Dependency Sequencing" section in D2 explicitly states "Error workflow must be created and set as default BEFORE timeout changes… timeouts without alerting reintroduce silence" and marks retry/credential/queue settings as independent
3	#1 "no error workflow" is silence-cause, not failure-cause — needs clarification	**RESOLVED**	D1 #1 now includes: "*Note: This explains why failures are invisible, not why they occur — it amplifies every other root cause into a 'silent' failure*"

Non-blocking notes from the prior report (defaults matching spec, Gmail "In production" insight, VPS scoping) remain intact in the revision — nothing was broken by the fixes.

Secondary check: The added 429 content introduced no new problems; the Retry-After guidance is UI/config-level and within scope (no custom node required).

Verdict: **PASS**

All three prior issues — including the two "required fixes" — are fully addressed. The deliverable now satisfies all eight acceptance criteria and the spec's edge-case table (OpenAI timeout vs. rate limit, dependency sequencing, category subtlety on #1) with no new defects introduced.

Part 4 · Client Handoff Note

Delivered: n8n silent failure diagnostic and hardening plan with 4 ranked root causes and fixes. Use it by checking each cause in order, starting with the confirmation check, then applying the fix. First verify Settings → Error Workflow is set and past failed executions exist without alerts. Then set execution/OpenAI timeouts, re-authorize Gmail OAuth, and monitor VPS memory for OOM kills. Success = failures trigger alerts, no stuck executions, credentials pass test, and n8n stays up.

Fix Application Log

#	Prior QA Issue	Resolution	Status
1	OpenAI rate limits (HTTP 429) not distinguished from timeouts	D1 #2 now includes a Timeout Hang vs. Rate Limit 429 comparison table, separate confirmation checks, and separate fixes (backoff/retry, `Retry-After`, error branch).	**Resolved**
2	D2 dependency sequencing was thin	Added a “Dependency Sequencing” section: error workflow must exist before timeout changes; retry/credential/queue settings are independent.	**Resolved**
3	#1 “no error workflow” framed as failure cause, not silence cause	D1 #1 now clarifies it explains why failures are invisible, not why they occur.	**Resolved**

Non-blocking notes from the prior report remain intact. No new defects introduced by the fixes.

Final QA Verdict: **PASS**